

Say, the current interest rate is i and an investment project that costs Rs. A , is expected to last for T periods. In each period it yields a return of R_t , where $t = 1, 2, \dots, T$. In this case, the present value of the stream of returns from the project P is calculated as:

$$P = \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \frac{R_3}{(1+i)^3} + \dots + \frac{R_T}{(1+i)^T} \quad \dots (2.2c)$$

If profit is the only motive of investment, this project will be taken up only if $(P - A) \geq 0$.

For example, if you are willing to lend Rs.100 at 10 per cent rate of interest for a period of 1 year, then at the end of the year you will receive Rs.110. So, at the current interest rate of 10 per cent, the *present discounted value* of Rs.110 is Rs. 100 [where, $100 = 110 / (1+0.1)$].

Now, for the economy as a whole, for any given rate of interest, i_0 , the corresponding level of investment I_0 by firms would consist of all the projects for which $P \geq A$. Now, if the interest rate *increases* to i_1 (R_t and A remaining unchanged), P would *decrease*. Consequently, some projects for which $P \geq A$, would now be unprofitable. At a higher interest rate, therefore, all the projects for which $P < A$ would not be taken up, so there would be a fall in the associated level of investment I_1 . Thus, when $i_0 > i_1$ we have $I_0 < I_1$.

If we include interest rate as an explanatory variable in the investment function, we have

$$I = c + dY - e i \quad \dots (2.2d)$$

where $c > 0$, $d > 0$ and $e > 0$. Here, e captures the interest responsiveness of planned investment expenditure.

Check Your Progress 1

- 1) Discuss the main features of the Keynesian consumption function.

.....

.....

.....

.....

.....

- 2) Through a diagram, explain how (i) a decrease in autonomous consumption, and (ii) an increase in the MPC, would affect the consumption function.

.....

.....

.....

.....

.....

- 3) Distinguish between autonomous investment and induced investment.

.....

.....

.....

.....

.....

- 4) State the reasons for the inverse relationship between interest rate and investment level.

.....

.....

.....

.....

.....

2.3 DETERMINATION OF OUTPUT IN THE KEYNESIAN MODEL

The Keynesian model is based on certain assumptions. First, there are unemployed resources in the economy. In other words, there is underutilized productive capacity (e.g., factories operating below capacity, unused machines, labour without employment, etc.). It follows from this that output in such an economy is demand determined, i.e., existing production capacity can be used to step up production in case there is an increase in aggregate demand for output. Second, money wages and prices are not flexible (i.e., there are price rigidity and wage rigidity). Since there is unemployment and workers are willing to work at the going wage rate, an increase in the supply of output is possible at the given price level. Third, output is assumed to be a homogenous commodity used both for consumption and investment.

We will present the simplest Keynesian model applicable to a three-sector economy consisting of households, firms and the government. The economy is closed (i.e., no external trade). Later on we relax this assumption and allow for external trade.

In the three-sector economy, aggregate demand comprises (i) planned consumption by households (C), (ii) planned investment by firms (I), and (iii) expenditure on goods and services by the government (G). Initially we assume that investment is fully autonomous, an assumption that is relaxed later. The government is assumed to raise taxes and here we consider the cases of both lump sum taxes as well as proportional taxes. Later we will also consider the implications that arise when the government makes transfer payments to households (e.g., pensions, unemployment dole, etc.).

Therefore, in our simple Keynesian model

$$\text{Aggregate Demand (AD)} = C + I + G$$

By substituting the values of C, I and G, we obtain

$$AD = a + b(Y - T) + \bar{I} + G \quad \dots (2.3)$$

Here T is a lump sum tax levied by the government; the investment function is given by (2.2a).

2.3.1 Determination of Equilibrium Output

The equilibrium condition in the Keynesian model is that aggregate demand for output is equal to aggregate supply (Y) or, the total output produced in the economy. Note that at equilibrium, the plans of households and firms (regarding expenditure on consumption and investment) are also fulfilled. In case plans are *not* fulfilled there would be changes in aggregate demand (as households and firms would adjust their plans) and hence output produced would change as well, so that the situation would *not* be one of equilibrium.

The equilibrium condition in the economy is achieved when

$$\text{Aggregate Supply} = \text{Aggregate Demand}$$

From equation (2.3) we know that

$$Y = a + b(Y - T) + \bar{I} + G$$

Solving this for Y, equilibrium output (Y_e) can be obtained as:

$$Y = a + bY - bT + \bar{I} + G$$

Or,

$$Y - bY = a - bT + \bar{I} + G$$

Thus,

$$Y_e = (a - bT + \bar{I} + G)/(1 - b) \quad \dots (2.4)$$

From (2.4) it should be clear to you that equilibrium output would be higher at higher levels of (i) autonomous investment, (ii) government expenditure, and (iii) the MPC. Recall that $0 < \text{MPC} < 1$, so that the term $(1 - b)$ in the denominator is always positive.

We will now carry out graphical analysis to further understand the process of attainment of equilibrium in the simple Keynesian model. First we will consider the case of a two-sector economy with only households and firms and no government. Later we will consider a three-sector economy which has the government sector as well.

Two-Sector Economy without Government

In a two-sector economy, there are only households and firms. Therefore, aggregate demand has only two components, C and I. Thus, $AD = C + \bar{I}$. Setting $G = T = 0$ in equation (2.4) above, we see that equilibrium output would be